AI-Driven Data Research in Quantitative Finance Corporate Project with WorldQuant Consulting (Beijing) Co., Ltd.

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Abstract

This project develops an end-to-end framework for converting SEC Item 1A "Risk Factors" disclosures into quantitative forecasting and portfolio-construction signals. Annual 10-K risk paragraphs are extracted from EDGAR filings, classified into a dynamically updated macro–meso risk taxonomy, aggregated into firm-year risk exposure vectors, and used to construct semantic peer networks. The forecasting model is a topology-only Spatio-Temporal Graph Attention Network (ST-GAT): cosine similarity between risk exposure vectors determines whether a peer edge exists, but raw cosine weights are not fed directly into the graph attention layer. This produces a strict ablation between the full risk graph, $\mathbf{A}_t = \mathbf{I} + \mathbf{A}_{risk,t}$, and an identity baseline, $\mathbf{A}_t = \mathbf{I}$, which uses only each equity's own financial features.

The forecasting panel covers 2006–2024, with chronological splitting into training years 2006–2016, validation years 2017–2020, and out-of-sample test years 2021–2024. Each firm-year node uses $F = 12$ backward-looking financial and market features, standardized using training-year statistics only, and predicts one-year-ahead July-to-June cumulative return and annualized volatility. The Meso-level ST-GAT provides the strongest OOS results, improving averaged RMSE from 0.3345 to 0.3001, averaged MAE from 0.2314 to 0.1991, and averaged Spearman rank correlation from -0.0139 to 0.1701 relative to the identity baseline. For volatility forecasting, the Meso ST-GAT reduces RMSE from 0.2240 to 0.1565 and improves Spearman rank from -0.1351 to 0.1770.

Graph diagnostics show that the Meso improvement is not driven by dense graph smoothing. In the 2021–2024 OOS period, the Meso topology retains only 38–118 off-diagonal directed edges per year, with active-node density below 0.14%, while the Macro topology retains 3,900–4,460 edges per year, with density around 5.1%–5.5%. This indicates that granular risk categories form sparse, high-confidence semantic peer links, whereas broad Macro categories create substantially denser and noisier connectivity.

Finally, the project extends the forecasting model into an institutional-style portfolio backtest. Annual ST-GAT predictions are held fixed over their July-to-June forecast window, while portfolios are rebalanced monthly using next-month execution timing, transaction costs of 10 bps, slippage of 5 bps, and SPY as benchmark. The strongest strategy ranks stocks by the Meso ST-GAT predicted return, holds the top quintile long and bottom quintile short, and achieves 7.93% annualized return, 7.47% annualized volatility, a Sharpe ratio of 1.06, Sortino ratio of 2.01, maximum drawdown of -6.90%, and beta of 0.24 against SPY. A top-minus-bottom diagnostic gives a statistically significant 17.69% annualized spread ($t = 2.18$, $p = 0.034$), compared with an insignificant 6.62% spread for the identity baseline return-only signal.

Keywords – Text classification; Large language models; SEC Item 1A; Risk taxonomy; Graph Neural Networks; ST-GAT; Portfolio backtesting; Volatility forecasting; Risk similarity; Semantic peer networks; Volatility contagion.

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1 INTRODUCTION

1.1 Background and Motivation

Since 2005, the U.S. SEC has required public companies to disclose material risks in Item 1A "Risk Factors" of Form 10-K. Prior empirical studies show that these risk-factor disclosures contain market-relevant information and shape investors' risk perceptions (Kravet and Muslu, 2013; Campbell et al., 2014; Hope et al., 2016). These narratives encode exposures to regulatory, operational, cyber, liquidity, and macroeconomic risks—information critical for valuation, portfolio construction, and systemic risk monitoring. However, the text is long, legalistic, and inconsistent across firms and years. Dyer et al. (2017) document that 10-K textual disclosure has expanded substantially over time and that boilerplate language has become increasingly common. Some firms substantially rewrite their risk profiles annually, while others exhibit exactly zero drift, copying boilerplate text. This unstructured format further hinders systematic analysis and motivates the need for structured NLP-based extraction.

1.2 Problem Statement

We address two core problems:

1. **Transformation:** How to reliably classify each risk paragraph into a consistent, interpretable taxonomy?
2. **Validation:** How to prove that the resulting quantitative signals have economic meaning, e.g., predict stock return or explain cross-sectional variation?

1.3 Research Questions

- **RQ1:** Can LLMs reliably classify risk disclosures into a hierarchical taxonomy?
- **RQ2:** Does a graph defined by textual risk similarity (ST-GAT) improve volatility and return prediction over an identity-matrix baseline that uses only each equity's own data?
- **RQ3:** Do ST-GAT forecasts translate into economically meaningful dynamic portfolio performance under monthly rebalancing, transaction costs, and benchmark comparison?

1.4 Contributions

- **LLM-driven hierarchical taxonomy with dynamic evolution:** From 2022 to 2024, precision improves from 94.7% to 98.5% by adding categories such as "Technology Adoption & Clinical Utility Risk" and reorganizing emergent themes.

- **Topology-only ST-GAT with strict spatial ablation:** The main forecasting model uses $\mathbf{A}_t = \mathbf{I}+\mathbf{A}_{risk,t}$, where off-diagonal edges are selected by thresholded cosine similarity of risk exposure vectors. The identity baseline uses $\mathbf{A}_t = \mathbf{I}$, isolating the incremental contribution of textual peer topology.

- **Macro-versus-Meso semantic resolution comparison:** Meso-level risk topology produces sparse, high-confidence semantic peer links and outperforms the identity baseline across return and volatility metrics. Macro topology is substantially denser and produces mixed results, improving return ranking but not volatility ranking.

- **Dynamic portfolio extension:** Annual ST-GAT forecasts are converted into monthly rebalanced long-only and long-short portfolios. The best Meso ST-GAT return-only long-short portfolio achieves a Sharpe ratio of 1.06, maximum drawdown of -6.90%, and statistically significant annualized top-minus-bottom spread of 17.69%.

- **Multi-model validation:** Integration of LLM taxonomy construction, ST-GAT forecasting, portfolio backtesting, EGARCH-X/DAV diagnostics, Fama-MacBeth regressions, SAR spillover analysis, and example-based validation.

1.5 Report Organization

The remainder of this report is organized as follows. Section 2 reviews financial textual analysis, SEC risk disclosure studies, graph neural networks, volatility modeling, and cross-sectional asset-pricing validation. Section 3 describes the data sources and preprocessing pipeline, including risk-factor extraction and the enhanced macro-financial feature matrix. Section 4 details the base-year taxonomy construction and LLM-driven dynamic taxonomy update mechanism. Section 5 formally defines the ST-GAT architecture, including graph construction, identity-matrix ablation, spatial attention, temporal convolution, and masked multi-task loss. Section 6 specifies the tensor construction, train-only standardization, active-mask logic, and feature-target alignment. Section 7 presents the chronological experimental design, Optuna tuning process, and hyperparameter configurations. Section 8 reports forecasting results and graph-density diagnostics. Section 9 introduces the dynamic portfolio backtesting extension, including monthly rebalancing, transaction costs, score ablations, and top-minus-bottom spread diagnostics. Sections 10 and 11 discuss limitations and conclude. Supplemental econometric baseline evaluations (DAV, Fama-MacBeth, and SAR models) and a representative sample of the evolved taxonomy are provided in the Appendices.

2 LITERATURE REVIEW

2.1 Textual Analysis in Finance: From Dictionaries to LLMs

Lexicon-based methods measure sentiment using pre-defined word lists but cannot capture context, negation, or industry-specific meanings (Loughran and McDonald, 2011; Li, 2010).

Topic modeling methods such as Latent Dirichlet Allocation discover latent themes but require a pre-specified number of topics and ignore hierarchical structure (Blei et al., 2003; Bao and Datta, 2014). Transformer-based LLMs overcome many of these limitations by learning bidirectional contextual representations (Devlin et al., 2019; Liu et al., 2019; Beltagy et al., 2020). Recent financial-text benchmarks and working papers further motivate domain-specific LLM evaluation on SEC filings (Choi et al., 2026; Jiang et al., 2026).

Gap 2.1: No prior work systematically compares multiple LLM architectures (DeepSeek-R1, GPT-4o-Mini, Llama-3.1) for hierarchical risk taxonomy classification of Item 1A, nor evaluates multi-level prompting strategies.

2.2 Empirical Studies of SEC Item 1A Risk Factor Disclosures

Prior accounting and finance studies establish that SEC Item 1A risk-factor disclosures contain market-relevant information. Kravet and Muslu (2013) find that changes in risk disclosures are associated with investors' risk perceptions, while Hope et al. (2016) show that more specific risk disclosures trigger stronger market responses. Campbell et al. (2014) document the information content of mandatory risk-factor disclosures, and Dyer et al. (2017) document the rising length, boilerplate, and stickiness of 10-K text over time. Grundy and Petry (2021) apply sentence-level LDA to S&P 1500 firms and show that quantified risk-factor text explains a meaningful fraction of cross-sectional risk variation. However, no prior study has systematically quantified year-over-year cosine drift in risk vectors or linked zero-drift boilerplate firms to the performance of graph-based predictive models.

Gap 2.2: The phenomenon of exactly zero drift (e.g., GE, NTAP) and its role in GNN underperformance has not been empirically established.

2.3 Graph Neural Networks (GNNs) in Financial Forecasting

Graph neural networks propagate information over non-Euclidean structures and have become a natural tool for modeling relational dependence (Veličković et al., 2018). Spatio-temporal graph attention architectures have also been applied to financial volatility forecasting and risk modeling by propagating information through graphs of interconnected assets (Kumar et al., 2024). Existing financial GNN studies typically construct graphs from industry classifications, historical correlations, supply-chain relations, or other static proxies. Niu et al. (2024) emphasize that static graph construction can fail to capture the evolving nature of financial relationships.

Gap 2.3: No prior study has constructed a graph directly from the semantic content of 10-K risk disclosures, systematically compared macro- versus meso-level graph resolutions, or performed a strict ablation study (replacing the graph with an identity matrix) to isolate the contribution of network topology.

2.4 Volatility Modeling with Textual Features

GARCH and EGARCH models capture volatility clustering and leverage effects in asset returns (Bollerslev, 1986; Nelson, 1991). Extensions such as GARCH-X and EGARCH-X allow exogenous variables to enter the conditional variance equation (Naimoli et al., 2023). A few studies have incorporated news-based sentiment or information volume, but the frequency mismatch between annual risk disclosures and daily volatility has received little attention.

Gap 2.4: No prior work systematically documents how annual risk scores, when repeated daily, are absorbed into the long-run variance intercept, nor provides a diagnostic framework (BIC, Diebold-Mariano, Mincer-Zarnowitz) to identify when textual features add value versus add noise.

2.5 Cross-Sectional Asset Pricing with Textual Features

The Fama-MacBeth two-step procedure is the standard methodology for testing whether a firm characteristic commands a cross-sectional risk premium (Fama and MacBeth, 1973). Gao (2016) proposes a text-implied risk factor using language-model embeddings, and later disclosure-risk factor studies extend traditional factor models with textual risk signals (Stevens Institute of Technology, 2020). Lopez-Lira (2019) also shows that textual risk factors can be informative for the cross-section of returns. However, these approaches have not been applied to LLM-derived hierarchical risk scores from Item 1A.

Gap 2.5: The Fama-MacBeth framework has never been used to test whether LLM-based risk scores from 10-K filings explain cross-sectional variation in returns after controlling for size, momentum, and trailing volatility.

2.6 Research Gap and the Present Contribution

Synthesizing the existing literature, four interconnected gaps remain: (i) the absence of quantitative metrics for year-over-year textual drift linked to predictive performance; (ii) the lack of semantic-similarity graph architectures with rigorous spatial ablation protocols; (iii) the absence of a diagnostic framework to address the frequency mismatch between annual disclosures and daily volatility; and (iv) a lack of Fama-MacBeth validation for LLM-derived risk scores from SEC Item 1A.

The present research addresses these gaps by integrating an LLM-driven dynamic taxonomy, a multi-resolution topology-only ST-GAT architecture, and a suite of econometric validation models. The main forecasting result is that the granular Meso-level topology outperforms the identity baseline across both return and volatility targets. In the 2021–2024 OOS period, the Meso ST-GAT improves averaged RMSE from 0.3345 to 0.3001 and averaged Spearman rank correlation from -0.0139 to 0.1701. For volatility specifically, the Meso ST-GAT reduces RMSE from 0.2240 to 0.1565 and improves Spearman rank from -0.1351 to 0.1770. The portfolio extension further demonstrates economic value: the Meso ST-GAT return-only long-short strategy achieves a Sharpe ratio of 1.06 and a statistically significant annualized top-minus-bottom spread of 17.69% ($p = 0.034$). Supplemental DAV, Fama-MacBeth, and SAR analyses are used as auxiliary validation exercises, while the central contribution is the strict identity-matrix ablation showing that granular textual-risk topology adds predictive and economic information beyond own-firm features alone.

3 DATA & PRE-PROCESSING

3.1 Data Sources

Our data pipeline draws from four primary sources:

- **SEC EDGAR Database:** The primary source of textual risk disclosures is the SEC's Electronic Data Gathering, Analysis, and Retrieval (EDGAR) system. We access filings through the SEC's RESTful API, which provides structured access to company filing histories.
- **CRSP-Compustat Merged (CCM):** We obtain accounting data (total assets, leverage, R&D expenditure, intangible assets) from Compustat and stock market data (returns, trading volume) from CRSP. The merged database allows firm-level linking of financial characteristics to 10-K filing dates.
- **Stooq API:** For high-frequency volatility estimation, we retrieve daily open-high-low-close-volume (OHLCV) data. This enables calculation of realized volatility at daily frequency.
- **Yahoo Finance / yfinance:** Daily adjusted prices and benchmark prices are used in the forecasting feature-generation and portfolio-backtesting modules, including the SPY benchmark equity curve. Stooq is retained for the supplemental DAV volatility experiment where OHLCV data are required.

3.2 Sample Description

All models in this study are trained and validated on the same unified dataset to ensure comparability across approaches. We construct a single, comprehensive sample that serves all modeling tasks.

Sample Construction. We expand the temporal scope to an 18-year longitudinal panel (2006-2024) and the cross-section to the dynamic S&P 500 universe. Due to strict EDGAR parsing heuristics (bypassing pre-2018 non-standard HTML and handling survivorship bias), the pipeline successfully embeds 2,387 pristine firm-year observations. This sample size is sufficient for:

- Training the ST-GAT model with adequate historical sequence data.
- Performing validation across diverse firm types and time periods.
- Constructing meaningful risk networks with sufficient node connectivity.

Sample Attributes.

- Time Period: 2006-2024 (18 years)
- Number of Firms: Dynamic S&P 500 universe
- Firm-Year Observations: 2,387
- Average Firms per Year: Approximately 132
- Total Risk Paragraphs Extracted: Approximately 35,000

Data Quality Assurance. To ensure data quality:

- Pre-2018 filings with non-standard HTML are bypassed due to parsing inconsistencies.
- Survivorship bias is partially mitigated where historical constituents and delisted tickers are available; a fully survivorship-bias-free reconstruction remains a limitation of the academic data environment.
- Duplicate filings (amendments, corrected versions) are identified by accession number and deduplicated, retaining only the most recent version.
- Irrecoverable filings are excluded from the sample and documented.

Market Effective Date Mapping. To avoid look-ahead bias, each risk feature is mapped to the first market close after public availability of the filing (typically 1–2 days after filing date). Returns on the filing date itself are not used as dependent variables. For the ST-GAT forecasting task, annual financial features are aligned to the July-to-June feature window and forward targets are measured over the following July-to-June window.

3.3 Data Pipeline: Four-Stage Extraction Process

The data pipeline consists of four sequential stages, each designed to progressively refine raw SEC filings into structured, analyzable risk disclosures.

Stage 1: Data Acquisition (Downloading HTML)
Objective: Download the complete annual reports (10-K forms) for a target list of major companies.
Source: SEC's EDGAR database at data.sec.gov.
Input: A curated list of company tickers (e.g., AAPL, MSFT, TSLA) and their corresponding

CIK numbers (unique identifiers used by the SEC).

Process: The script sends an HTTP request to the SEC API for each company's submissions history; finds the most recent filing where the form type is 10-K; extracts the URL for the primary HTML document; downloads the entire HTML content.

Output: Raw HTML files saved locally (e.g., AAPL_10K.html).

Stage 2: Data Extraction (Isolating "Item 1A")

Objective: Extract only the "Item 1A: Risk Factors" section from the large 10-K HTML files.

Why this is needed: A 10-K filing is very large; we need to discard irrelevant sections.

Process: Read HTML using an HTML parser; find boundaries with patterns like "Item 1A Risk Factors"; remove Table of Contents references; locate end point (next item, usually "Item 1B"); extract HTML elements between boundaries.

Output: New HTML files containing only the Risk Factors section.

Stage 3: Data Structuring (Separating Headings from Descriptions)

Step 3.1: Structural Formatting Analysis. Parse the HTML structure; identify bold text, italic/underlined text, and regular text. Group consecutive bold text as potential Risk Title, following regular text as Risk Description.

Step 3.2: Frequency-Based Validation (Format Frequency Heuristic). Filter false positives (section numbers, introductory phrases, legal qualifiers, boilerplate) by analyzing frequency of each formatting type across the document. Core principle: frequency inversely indicates hierarchical level—bolded text appears least frequently (main headings), bold-italic or underlined appears moderately (subheadings), regular text appears most frequently (descriptions). Implementation: count formatting types; classify lowest frequency as main headings, medium as subheadings, highest as descriptions; filter identical repeated phrases; cross-validate with position.

Stage 4: Data Export (Creating CSV Files)

Iterate files, parse and clean text, flatten hierarchical data into a table, aggregate into a master CSV. Output includes individual CSV files per company, a master CSV with all risks, and summary reports. Final columns: Filename, Company Ticker, Year, Risk Heading, Paragraph Text.

3.4 Macro-Financial Feature Space

The final ST-GAT feature matrix contains $F = 12$ node-level inputs. Firm-level features are computed over the July-to-June feature window associated with each risk year, while market-wide overlays are computed over the same window and shared across firms in that year.

Idiosyncratic Metrics (Firm-Specific):

- Annualized realized volatility over the feature window (Vol).
- Annualized downside volatility computed from negative daily log returns (Downside_Vol).
- Price deviation from the trailing 200-day moving average (MA_200_Ratio).
- Eleven-month momentum, excluding the most recent month (Momentum_11M).
- Average dollar trading volume and its log transform (Dollar_Volume and Log_Dollar_Volume).

Market and Macro Proxies:

- S&P 500 cumulative return and realized volatility (SP500_Ret, SP500_Vol).
- Average VIX level and VIX change over the feature window (VIX_Avg, VIX_Change).
- Average 10-year Treasury yield proxy and yield change (TenY_Yield_Avg, TenY_Yield_Change).

Forward Targets. The target variables are one-period-ahead cumulative return and annualized volatility over the following July-to-June window. Thus, for risk year t, input features and risk topology are observed over July t to June $t+1$, while Target_Ret and Target_Vol are measured over July $t+1$ to June $t+2$.

Feature Standardization. To prevent look-ahead bias, all node features are standardized using the training split only. Means, standard deviations, and median-imputation values are fitted on 2006–2016 data and then applied unchanged to validation and test regimes.

4 CONSTRUCTION OF BASE YEAR TAXONOMY AND DYNAMIC UPDATE TAXONOMY

4.1 Conceptual Framework of the Dynamic Taxonomy

The core innovation of this research lies in the transition from static, industry-defined risk classifications to a Dynamic Taxonomy. Traditional classification models frequently suffer from "semantic decay," a phenomenon where emergent risks—such as the rapid rise of generative Artificial Intelligence (AI) or evolving Environmental, Social and Governance (ESG) mandates—are misclassified into broad, outdated categories that fail to capture their specific implications. Furthermore, traditional systems often require intensive manual intervention from financial experts to define and update default risk categories, a process that is both time-consuming and prone to subjective bias.

To address this, we propose a dynamic taxonomy with two components: (i) a base taxonomy constructed from a reference year (2006) using unsupervised hierarchical clustering, and (ii) an evolutionary update mechanism that processes new annual data to add, merge, or reorganize categories when semantic deviation exceeds a threshold ($\theta = 0.45$ cosine similarity). This framework ensures the taxonomy remains data-driven without manual intervention.

4.2 Base Year Taxonomy Construction

The base taxonomy is built via a multi-stage pipeline, as detailed below.

4.2.1 Contextual Encoding

Each risk factor is represented by concatenating its Headings, Subheadings, and Raw Text fields into a full_context string. This preserves the structural hierarchy of the 10-K disclosure. We then generate 384-dimensional embeddings using the all-MiniLM-L6-v2 sentence transformer, following the sentence-embedding approach introduced by Reimers and Gurevych (2019). This model balances computational efficiency with semantic sensitivity.

4.2.2 Two-Stage Hierarchical Clustering

Macro-level clustering:

- Dimensionality Reduction: UMAP with n_neighbors=15, min_dist=0.1, and n_components=5 is utilized to compress the 384-dimension embeddings into a lower-dimensional manifold (5 components). UMAP is designed to preserve local manifold structure, which is essential for effective density detection (McInnes et al., 2018).
- Density-based clustering: HDBSCAN with min_cluster_size=10 and min_samples=5 is applied to identify clusters of varying densities and shapes (Campello et al., 2013).
- This yields macro risk categories (e.g., "Supply Chain Risk", "Regulatory Compliance").

Meso-level clustering (within each macro cluster):

- For each macro cluster, we extract its embeddings and apply UMAP again (n_neighbors=8, min_dist=0.2, n_components=8).
- HDBSCAN with min_cluster_size=5 and min_samples=3 identifies meso subcategories.
- Meso categories are specific risk themes (e.g., "Cross-Border Data Transfer", "Technology Standardization").

4.2.3 Refined Hybrid Centroids

For each discovered category (macro or meso), we compute a hybrid centroid that combines data-driven and semantic signals:

$$\mathbf{C}_{\text{refined}} = 0.7 \times \mathbf{X}_{\text{mean}} + 0.3 \times \mathbf{X}_{\text{LLM}} \tag{1}$$

where:

- $\mathbf{X}_{\text{mean}}$ represents the arithmetic mean of all embeddings assigned to the category.
- $\mathbf{X}_{\text{LLM}}$ represents the embedding of a Centroid Summary generated by an LLM (DeepSeek) from a sample of category members. The prompt asks the LLM to produce a 50-word synthesis of the category's core "DNA".

The weights (0.7 / 0.3) are heuristic; they prioritize empirical data while anchoring the centroid to human-interpretable semantics, preventing drift caused by outliers.

4.2.4 Handling Unclassified ("Noise") Points

After meso clustering, any risk factor not assigned to a meso category is force-assigned to the nearest meso centroid using cosine similarity. This ensures 100% coverage for downstream risk exposure vector construction.

4.3 Dynamic Taxonomy Update (Year-over-Year)

The evolutionary pipeline processes new-year data (e.g., 2007) and updates the taxonomy from the previous year (e.g., 2006). The workflow consists of four steps.

4.3.1 Initial Classification and Deviation Detection

Each new risk factor is embedded and compared to all existing meso centroids from the base taxonomy. The highest cosine similarity is recorded. If that similarity falls below a predefined threshold $\theta = 0.45$, the factor is flagged as a semantic deviation. The threshold was chosen empirically based on the 5th percentile of intra-category similarities in the training set.

4.3.2 Clustering Deviations

All deviation points are collected. If the number of deviations is at least MIN_CLUSTER_SIZE_MESO = 5, they are projected via UMAP and clustered with HDBSCAN. Each resulting cluster represents a candidate new risk theme.

4.3.3 LLM-Driven Evolution Decision

For each deviation cluster, we provide the LLM with:

- Sample texts from the cluster,

- The top-3 closest existing categories (by cosine similarity),
- A summary of the current taxonomy.

The LLM chooses one of three actions (coded as A, B, C in the prompt):

Table 1: Decision types for taxonomy evolution

Decision	Action	Description
A	ADD	Create a new meso category. The LLM provides a recommended name and parent macro category.
B	MERGE	Absorb the deviation cluster into the closest existing category. The existing centroid is updated via an exponential moving average (80% old, 20% new) to reflect drift.
C	REORGANIZE	Move an existing meso category to a different macro parent. The LLM specifies the target category name and the new parent macro.

4.3.4 Execution and Persistence

- **ADD:** A new meso category is appended to the taxonomy with a refined hybrid centroid ($0.7\times$ math centroid of the cluster $+0.3\times$ embedding of the LLM-recommended name). All deviation points in that cluster are labeled with the new category.
- **MERGE:** The closest existing category's centroid is updated to $0.8\times$ old_centroid $+0.2\times$ cluster_math_centroid, preserving stability while adapting to new data.
- **REORGANIZE:** The parent_macro field of the target category is updated.

4.4 Empirical Validation and Findings

To evaluate the effectiveness of the proposed dynamic taxonomy, we compare the base taxonomy (built from 2022 data) against the evolved taxonomy (after processing 2024 data). Two validation approaches are used: (i) a quantitative coverage analysis measuring the reduction of ambiguous classifications, and (ii) a qualitative firm-level case study illustrating how risk profiles shift over time.

4.4.1 Coverage Improvement: Baseline vs. Evolved Taxonomy

For each risk factor in a given year, we compute its cosine similarity to the closest meso-category centroid in the taxonomy. A factor is flagged as "ambiguous/outlier" if its maximum similarity falls below the deviation threshold $\theta = 0.45$ (as defined in Section 4.3.1). The percentage of precisely classified factors (i.e., similarity ≥ 0.45) serves as a direct measure of taxonomic coverage.

- **2022 Baseline (fixed taxonomy):** Only 94.7% of risk factors achieved a similarity score ≥ 0.45. The remaining 5.3% were classified as ambiguous, indicating that the base taxonomy, while internally consistent, already contained gaps relative to the full diversity of risk disclosures.
- **2024 Evolved (after applying ADD/MERGE/REORGANIZE):** The coverage increased to 98.5%. The reduction of ambiguous factors from 5.3% to 1.5% demonstrates that the evolutionary mechanism successfully absorbed previously unseen risk patterns, either by creating new categories (ADD) or by merging deviations into existing ones (MERGE).

The remaining 1.5% of factors were isolated deviations that did not form clusters of size ≥ 5 (see Section 4.3.2). These were force-assigned to the nearest centroid, guaranteeing full coverage for downstream risk vector construction but potentially introducing minor classification noise.

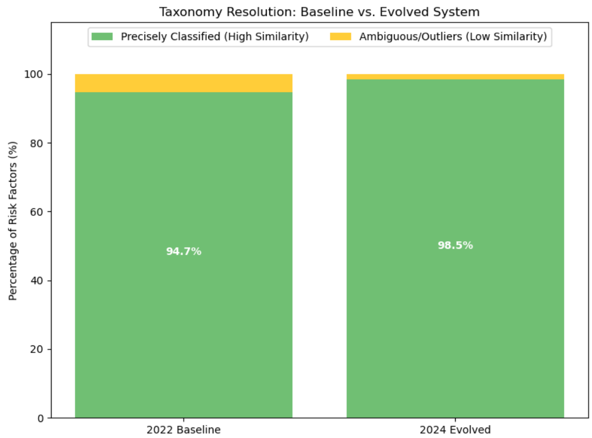

Figure 1: Taxonomy coverage improvement – Baseline (2022) vs. Evolved (2024).

4.4.2 Firm-Level Risk Profile Evolution: Case Study

Beyond aggregate metrics, the dynamic taxonomy enables tracking of individual firms' risk exposure vectors over time. We illustrate this using Apple Inc. (AAPL) as a representative case. For each year, we construct a risk exposure vector for AAPL by normalising the count of risk factors assigned to each meso category by the firm's total number of risk factors. The resulting vectors for 2022 (using the base taxonomy) and 2024 (using the evolved taxonomy) are compared via a radar chart.

- **2022 Profile:** Apple's risks were predominantly mapped to broad categories such as Supply Chain Risk, Geopolitical Risk, and General Technology Risk. These labels reflect the base taxonomy's limited granularity.

- **2024 Profile:** After evolution, the taxonomy had acquired specialised meso categories including Cross-Border Data Transfer & Localization, Technology Standardization Risk, and Environmental Compliance Risk. Apple's 2024 exposure vector shows a clear migration toward these emergent nodes, with a corresponding decrease in the weight of generic categories.

This shift is not manually imposed; it emerges automatically from the evolutionary pipeline. When Apple's 2024 risk factors were embedded and found to be distant from existing centroids, the LLM decided to ADD new categories that precisely matched the language of Apple's disclosures (e.g., references to "data localisation requirements" and "proprietary technology standards"). The radar chart visually confirms that the evolved taxonomy captures the firm's strategic pivot from broad supply chain concerns to technology-specific regulatory risks.

4.4.3 Discussion of Validation

The coverage improvement (5.3% $\rightarrow$ 1.5% ambiguous) and the firm-level case together provide empirical evidence that:

- Semantic decay is real – a fixed taxonomy loses accuracy over a two-year horizon.
- The evolutionary mechanism mitigates decay – by detecting deviation clusters and applying ADD/MERGE/REORGANISE decisions, the taxonomy adapts without human intervention.
- The resulting risk exposure vectors are more informative – firms' risk profiles become sparser and more specialised, reflecting actual changes in disclosure content.

Limitations: The validation is limited to two time points (2022 and 2024). A full longitudinal study (e.g., rolling updates from 2015 to 2024) would be required to assess long-term stability. Additionally, the coverage metric relies on the fixed similarity threshold (0.45); sensitivity analysis on this threshold is left for future work.

5 MATHEMATICAL FORMULATION OF THE SPATIO-TEMPORAL ARCHITECTURE

The predictive objective of the ST-GAT model is to determine whether semantic similarity in annual SEC Item 1A risk disclosures provides cross-sectional information beyond each firm's own financial trajectory. The architecture is therefore designed as a strict spatial ablation. The full model uses a dynamic risk-similarity topology, while the baseline uses only self-loops. This section defines the graph, attention mechanism, temporal convolution layer, masking logic, and multi-task loss used for one-year-ahead return and volatility prediction.

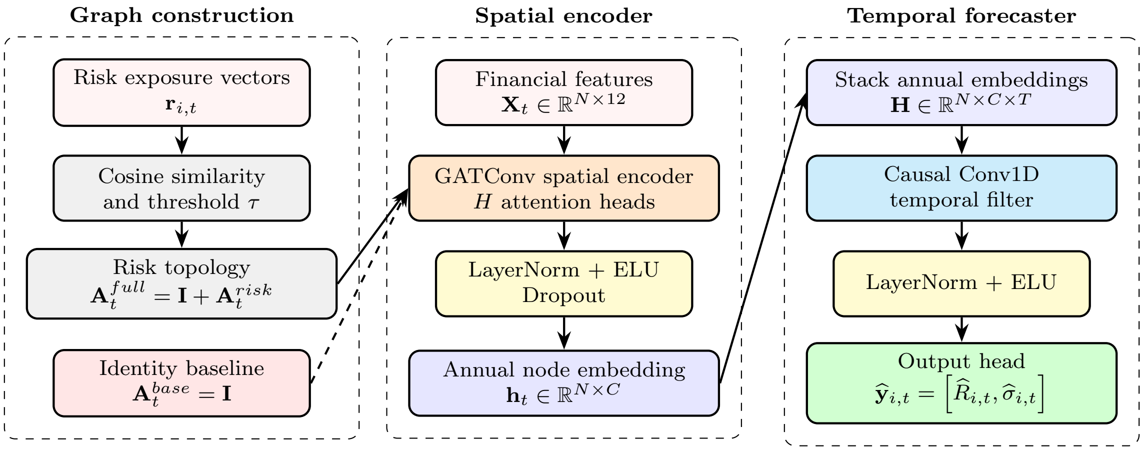

Figure 2: ST-GAT block architecture and identity-matrix ablation. Risk exposure vectors define the sparse graph topology, financial node features enter the spatial GAT encoder, and annual embeddings are passed through a causal temporal forecaster to predict one-year-ahead return and volatility. The full model uses textual peer topology, whereas the identity baseline removes all off-diagonal links. Cosine similarity determines edge existence, while attention weights are learned by GATConv.

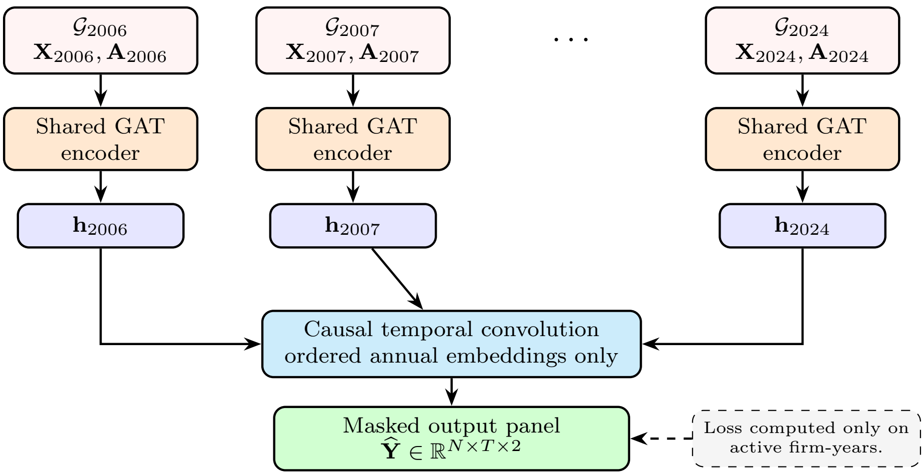

Figure 3: Temporal processing of annual graph snapshots. The shared GAT encoder is applied to each annual graph, and a causal temporal convolution produces masked return-volatility predictions across the full panel.

5.1 Dynamic Risk-Similarity Graph Definition

For each year t, the financial universe is represented as an attributed graph:

$$\mathcal{G}_t = (\mathcal{V}, \mathcal{E}_t, \mathbf{A}_t, \mathbf{X}_t), \tag{2}$$

where $\mathcal{V} = \{v_1, \ldots, v_N\}$ is the global node universe, $\mathcal{E}_t$ is the annual edge set, $\mathbf{A}_t \in \mathbb{R}^{N\times N}$ is the adjacency matrix, and $\mathbf{X}_t \in \mathbb{R}^{N\times F}$ is the node feature matrix. In the final implementation, $F = 12$ financial and macro-market features are used for each firm-year.

For each firm i and year t, the risk exposure vector $\mathbf{r}_{i,t}$ is constructed from normalized paragraph shares:

$$r_{i,k,t} = \frac{\text{number of Item 1A paragraphs of firm } i \text{ assigned to risk category } k \text{ in year } t}{\text{total Item 1A paragraphs of firm } i \text{ in year } t}. \tag{3}$$

Macro graphs use first-level risk categories, while Meso graphs use second-level risk categories. The off-diagonal semantic similarity between firms i and j is computed using cosine similarity:

$$s_{ij,t} = \frac{\mathbf{r}_{i,t}^{\top}\mathbf{r}_{j,t}}{\|\mathbf{r}_{i,t}\|_2\|\mathbf{r}_{j,t}\|_2}. \tag{4}$$

A risk edge is retained only if similarity exceeds an optimizable threshold τ:

$$A_{ij,t}^{risk} = \begin{cases} 1, & i \neq j \text{ and } s_{ij,t} \geq \tau, \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

The main reported specification is topology-only: cosine similarity determines whether an edge exists, but raw cosine weights are not directly passed into the GAT attention mechanism. This design was selected after weighted-edge attention was found to be less stable empirically. Edge weights are retained for diagnostics and optional ablations, but the primary model allows the GAT layer to learn propagation weights from node features and graph topology.

5.2 Full Graph versus Identity Baseline

The final spatial ablation is defined as:

$$\mathbf{A}_t^{full} = \mathbf{I}_N + \mathbf{A}_t^{risk}, \tag{6}$$

while the baseline is:

$$\mathbf{A}_t^{base} = \mathbf{I}_N. \tag{7}$$

Therefore, the full ST-GAT receives each firm's own information and information from textually similar peers, while the baseline receives only each firm's own information. This establishes a clean test of whether the NLP-derived risk topology contributes predictive information beyond isolated firm-level features.

Table 2: Adjacency structures used in the ST-GAT ablation.

Model	**Adjacency**	**Interpretation**
Full ST-GAT	$\mathbf{A}_t^{full} = \mathbf{I}_N + \mathbf{A}_t^{risk}$	Own-firm information plus peer spillovers from firms with similar 10-K risk exposure vectors.
Identity Baseline	$\mathbf{A}_t^{base} = \mathbf{I}_N$	Own-firm information only; all off-diagonal textual peer links are removed.

5.3 Spatial Graph Attention Mechanism

For each annual graph snapshot, a graph attention layer transforms node features into spatial embeddings, following the attention-based neighborhood aggregation principle of Graph Attention Networks (Veličković et al., 2018).

$$e_{ij,t} = \text{LeakyReLU}\left(\mathbf{a}^\top [\mathbf{W}\mathbf{x}_{i,t} \parallel \mathbf{W}\mathbf{x}_{j,t}]\right), \tag{8}$$

where $\mathbf{W}$ is a learnable projection, $\mathbf{a}$ is the attentional vector, and $\parallel$ denotes concatenation. The normalized attention weight is:

$$\alpha_{ij,t} = \frac{\exp(e_{ij,t})}{\sum_{k \in \mathcal{N}_t(i)} \exp(e_{ik,t})}. \tag{9}$$

The spatial embedding for node i is then:

$$\mathbf{h}_{i,t} = \sigma\left(\sum_{j \in \mathcal{N}_t(i)} \alpha_{ij,t} \mathbf{W}\mathbf{x}_{j,t}\right). \tag{10}$$

Multi-head attention is used to capture distinct modes of peer interaction:

$$\mathbf{h}'_{i,t} = \Big\Vert_{m=1}^{H} \sigma\left(\sum_{j \in \mathcal{N}_t(i)} \alpha_{ij,t}^{(m)} \mathbf{W}^{(m)} \mathbf{x}_{j,t}\right). \tag{11}$$

In the implementation, the GAT output is followed by layer normalization, ELU activation, and dropout. Self-loops are supplied explicitly by the data pipeline, so the graph convolution layer is configured without automatic self-loop insertion.

5.4 Temporal Convolutional Mechanism

After spatial encoding, annual embeddings are stacked into a temporal tensor:

$$\mathbf{H} \in \mathbb{R}^{N \times C \times T}, \tag{12}$$

where C is the concatenated spatial embedding dimension and T is the number of annual snapshots. A causal one-dimensional convolution is then applied along the temporal dimension, following the standard temporal-convolution principle that predictions at time t should not depend on future states (Bai et al., 2018):

$$\mathbf{z}_{i,t} = \sigma\left(\sum_{k=0}^{K-1} \mathbf{W}_T^{(k)} \mathbf{h}'_{i,t-k} + \mathbf{b}_T\right). \tag{13}$$

The causal truncation ensures that the representation at year t depends only on current and historical graph states. The final output layer maps the temporal representation to two forward targets:

$$\widehat{\mathbf{y}}_{i,t} = \left[\widehat{R}_{i,t}, \widehat{\sigma}_{i,t}\right]. \tag{14}$$

5.5 Masked Multi-Task Loss

Because the global universe contains firms that are not active in every year, the model uses a target-aware active mask $M_{t,i} \in \{0, 1\}$. A firm-year is active only when it has valid risk exposure data, valid financial features, and non-missing forward return and volatility labels. The training objective is the masked Huber loss averaged over active firm-years. Huber loss is used because it is less sensitive to extreme return observations than squared-error loss while remaining differentiable around zero (Huber, 1964):

$$\mathcal{L} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \frac{1}{\sum_i M_{t,i}} \sum_{i=1}^{N} M_{t,i}\, \ell_\delta\left(\hat{\mathbf{y}}_{i,t}, \mathbf{y}_{i,t}\right), \tag{15}$$

where ℓ_δ is the Huber loss with $\delta = 1.0$. The first output dimension $\widehat{R}_{i,t}$ denotes predicted one-year-ahead cumulative return, and the second output dimension $\widehat{\sigma}_{i,t}$ denotes predicted one-year-ahead annualized volatility. Return and volatility are optimized jointly with equal weight, while evaluation reports target-specific RMSE, MAE, and Spearman rank correlation.

6 DATA PIPELINE AND MULTI-DIMENSIONAL TENSOR STRUCTURE

The ST-GAT pipeline converts annual risk-score matrices and financial features into a sequence of graph snapshots. Each snapshot contains node features, sparse graph edges, optional edge attributes, forward targets, and an active mask. This section describes the tensor construction and the procedures used to avoid look-ahead bias.

6.1 Node Feature Engineering and Train-Only Standardization

For each node v_i and year t, the feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^{12}$ consists of six firm-level features and six market or macro proxy features. The twelve inputs are Vol, Downside_Vol, MA_200_Ratio, Momentum_11M, Dollar_Volume, Log_Dollar_Volume, SP500_Ret, SP500_Vol, VIX_Avg, VIX_Change, TenY_Yield_Avg, and TenY_Yield_Change.

Feature preprocessing is fitted only on the training split. Training-year medians are used to impute missing active-node features, and training-year means and standard deviations are used for standardization. The same fitted statistics are then applied to validation and OOS test years. This prevents information from 2017 onward from entering the feature scaling process.

Table 3: Final ST-GAT node feature set ($F = 12$).

Feature Category	Specific Metrics	Frequency
Firm-specific price and risk	Vol, Downside_Vol, MA_200_Ratio, Momentum_11M	Annual window
Firm-specific liquidity	Dollar_Volume, Log_Dollar_Volume	Annual window
Market regime	SP500_Ret, SP500_Vol, VIX_Avg, VIX_Change	Annual window
Interest-rate proxy	TenY_Yield_Avg, TenY_Yield_Change	Annual window

6.2 Feature-Target Alignment

The data pipeline follows a July-to-June alignment. For risk year t, financial and market features are measured from July 1 of year t to June 30 of year $t+1$. Forward targets are measured from July 1 of year $t+1$ to June 30 of year $t+2$. This matches the annual disclosure cycle while ensuring that the prediction target is strictly forward-looking.

6.3 Graph Snapshot Tensor Structure

For each year t, the graph builder produces:

$$\mathbf{X}_t \in \mathbb{R}^{N\times F}, \tag{16}$$

$$\textbf{edge\_index}_t \in \mathbb{N}^{2\times E_t}, \tag{17}$$

$$\textbf{edge\_attr}_t \in \mathbb{R}^{E_t}, \tag{18}$$

$$\mathbf{Y}_t \in \mathbb{R}^{N\times 2}, \tag{19}$$

$$\mathbf{M}_t \in \{0,1\}^{N}. \tag{20}$$

The complete supervised panel is then represented as:

$$\mathbf{Y} \in \mathbb{R}^{N\times T\times 2}, \qquad \mathbf{M} \in \{0,1\}^{T\times N}. \tag{21}$$

Unlike a random mini-batch design, training is performed on the full chronological graph panel with masked losses. This is appropriate because the graph topology and temporal convolution depend on ordered annual snapshots.

7 EXPERIMENTAL DESIGN AND ALGORITHMIC TUNING

The empirical design evaluates whether textual risk topology improves one-year-ahead return and volatility prediction under strict chronological splitting. Random cross-sectional splits are avoided because they would leak future macro regimes into model selection.

7.1 Strict Chronological Data Splitting

The sample is split into three non-overlapping temporal regimes:

- **Training Set (2006–2016):** Used to fit model weights, feature standardization statistics, and median imputation values.
- **Validation Set (2017–2020):** Used for Optuna hyperparameter selection and early stopping.
- **Out-of-Sample Test Set (2021–2024):** Used only for final evaluation.

7.2 Dual-Level Spatial Ablation Framework

Three graph settings are evaluated:

1. **Macro-Level Topology:** Edges are constructed from broad first-level risk category exposure vectors.
2. **Meso-Level Topology:** Edges are constructed from granular second-level risk category exposure vectors.
3. **Identity Baseline:** All off-diagonal risk edges are removed, leaving only self-loops.

This design tests both whether textual peer topology improves forecasting and whether semantic resolution affects the quality of that topology.

7.3 Algorithmic Tuning and Optimization

Macro and Meso graphs have different densities and information bandwidths, so they are tuned independently. We use Optuna with a Tree-structured Parzen Estimator sampler and a Median Pruner, following modern automated hyperparameter optimization practice (Akiba et al., 2019). Each study runs 50 trials. The objective minimizes validation Huber loss over 2017–2020, with maximum 200 epochs per trial, patience of 25 epochs, gradient clipping at 1.0, and early pruning of weak trials. The final forecast models are trained with maximum 300 epochs, patience of 40 epochs, and restoration of the best validation checkpoint.

Table 4: Optuna search space and final hyperparameters.

Hyperparameter	**Search Space**	**Macro ST-GAT**	**Meso ST-GAT**
Similarity Threshold (τ)	Uniform: $[0.75, 0.95]$	0.8128	0.8292
Attention Heads (H)	Categorical: $\{1, 2, 4, 8\}$	1	2
Spatial Dimension	Categorical: $\{8, 16, 32\}$	16	8
Dropout Rate (p)	Uniform: $[0.10, 0.50]$	0.3688	0.4979
Learning Rate (η)	Log-uniform: $[10^{-4}, 10^{-2}]$	6.69×10^{-3}	8.08×10^{-3}
Weight Decay	Log-uniform: $[10^{-6}, 10^{-3}]$	1.98×10^{-5}	5.83×10^{-5}
Best Validation Huber Loss	–	0.0378	0.0347

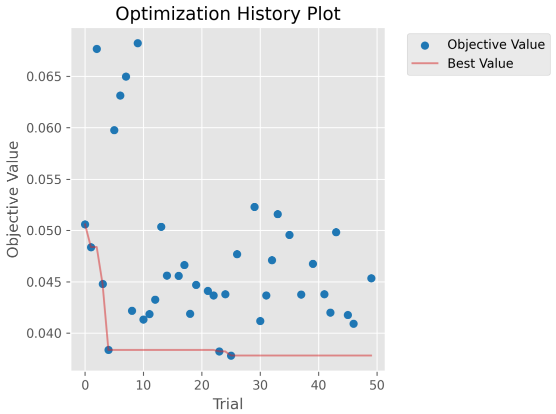

Figure 4: Optimization History (Macro Level)

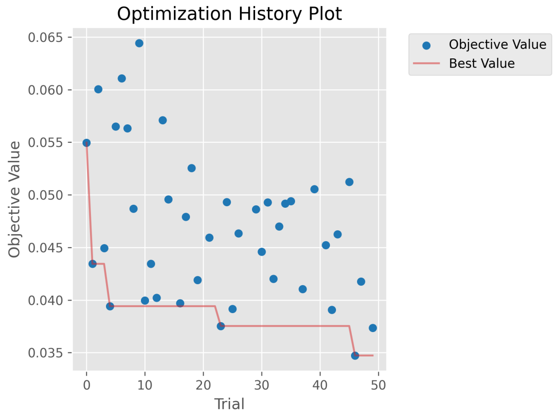

Figure 5: Optimization History (Meso Level)

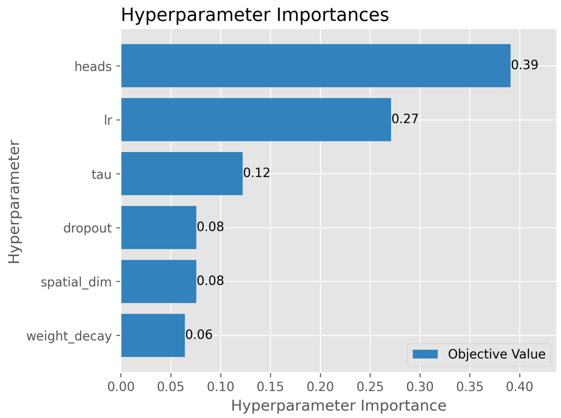

Figure 6: Hyperparameter Importance (Macro Level)

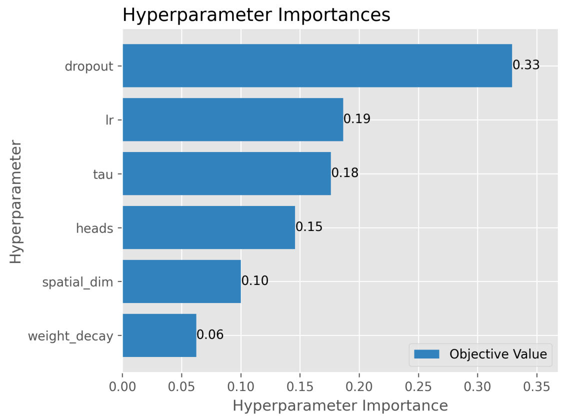

Figure 7: Hyperparameter Importance (Meso Level)

8 OUT-OF-SAMPLE PERFORMANCE AND EMPIRICAL RESULTS

This section reports OOS predictive performance on 2021–2024. Performance is measured by RMSE, MAE, and Spearman rank correlation. Spearman rank is particularly important because the portfolio extension relies on cross-sectional ranking rather than point forecasts alone.

8.1 Neural Network Convergence Dynamics

The final training stage uses validation early stopping and best-checkpoint restoration. Both full ST-GAT models and identity baselines are trained under the same chronological split, masked Huber loss, and gradient clipping. The convergence plots show training and validation Huber loss for the Macro and Meso experiments.

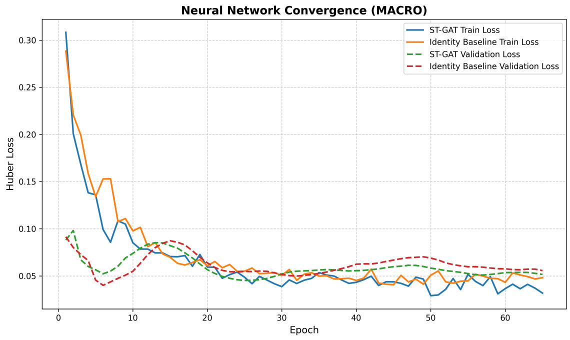

Figure 8: Loss Convergence (Macro Experiment)

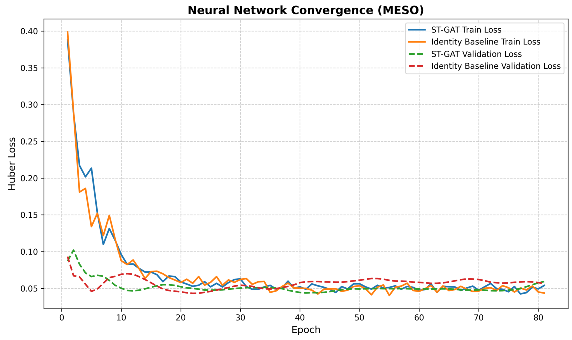

Figure 9: Loss Convergence (Meso Experiment)

8.2 Experiment 1: Macro-Level Systemic Risk Topology

Table 5: OOS Performance: Macro-Level ST-GAT vs. Identity Baseline (2021–2024)

Target	**Model**	**RMSE** (↓)	**MAE** (↓)	**Spearman** (↑)
Returns	ST-GAT	**0.4469**	**0.2700**	**0.0873**
Returns	Identity Baseline	0.4557	0.2801	-0.0396
Volatility	ST-GAT	0.1984	0.1653	-0.0356
Volatility	Identity Baseline	**0.1973**	**0.1627**	**0.0121**
Averaged	**ST-GAT**	**0.3227**	**0.2177**	**0.0258**
Averaged	Identity Baseline	0.3265	0.2214	-0.0138

The Macro graph improves return ranking relative to the identity baseline but is weaker for volatility. This suggests that broad first-level risk categories capture some market-wide return information but may over-connect firms when the target is future volatility.

8.3 Experiment 2: Meso-Level Granular Risk Topology

Table 6: OOS Performance: Meso-Level ST-GAT vs. Identity Baseline (2021–2024)

Target	**Model**	**RMSE** (↓)	**MAE** (↓)	**Spearman** (↑)
Returns	ST-GAT	**0.4437**	**0.2661**	**0.1631**
Returns	Identity Baseline	0.4450	0.2719	0.1072
Volatility	ST-GAT	**0.1565**	**0.1321**	**0.1770**
Volatility	Identity Baseline	0.2240	0.1909	-0.1351
Averaged	**ST-GAT**	**0.3001**	**0.1991**	**0.1701**
Averaged	Identity Baseline	0.3345	0.2314	-0.0139

The Meso topology outperforms the identity baseline across every target and metric. The largest improvement appears in volatility forecasting, where the ST-GAT reduces RMSE from 0.2240 to 0.1565 and reverses Spearman rank from -0.1351 to 0.1770. This indicates that granular textual peer links contain predictive information not captured by each equity's own financial data alone.

8.4 Graph Density Diagnostics: Macro versus Meso Topology

Graph diagnostics explain why semantic resolution matters. In the OOS period, the Macro graph is much denser than the Meso graph, creating thousands of off-diagonal links per year. The Meso graph instead retains a small number of high-confidence links.

Table 7: OOS graph-density comparison between Macro and Meso risk topologies.

Year	**Macro Edges**	**Macro Density**	**Meso Edges**	**Meso Density**
2021	3,900	5.45%	38	0.053%
2022	4,282	5.33%	82	0.102%
2023	4,450	5.17%	74	0.086%
2024	4,460	5.14%	118	0.136%

The Macro graph contains roughly 38–103 times more off-diagonal directed edges than the Meso graph during the OOS window. Combined with the forecasting results, this supports the interpretation that broad Macro categories may over-propagate generic disclosure noise, while granular Meso categories form a sparse, high-confidence semantic peer network.

9 DYNAMIC PORTFOLIO BACKTESTING EXTENSION

The forecasting results establish that the Meso topology-only ST-GAT improves cross-sectional return and volatility prediction relative to the identity baseline. However, a statistically better forecast does not automatically imply a tradable investment strategy. To evaluate whether the model produces economically meaningful signals, we extend the forecasting output into a monthly rebalanced portfolio backtest. This section formalizes the signal timing, portfolio construction rules, transaction-cost assumptions, score ablations, benchmark comparison, and top-minus-bottom spread diagnostic.

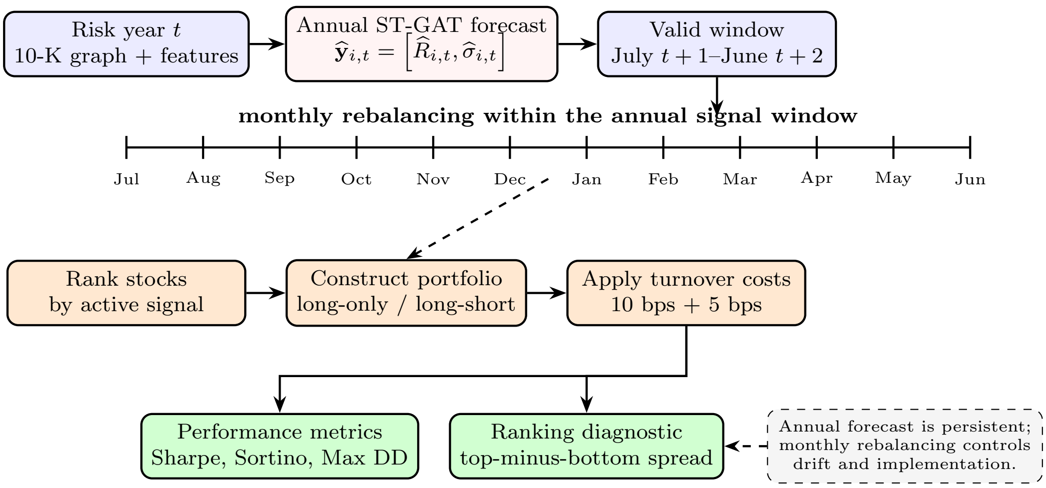

Figure 10: Portfolio signal timing. Annual ST-GAT predictions remain active over the July-to-June forecast window, while monthly rebalancing controls weight drift, turnover, and implementation costs.

9.1 Objective and Signal Choice

The portfolio extension is designed to answer whether the ST-GAT prediction panel can separate future winners from losers after accounting for realistic implementation frictions. The primary signal source is the Meso ST-GAT output because it is the strongest forecasting specification in Section 8. The identity-baseline predictions are retained as a portfolio benchmark, allowing the analysis to test whether textual peer topology improves allocation quality, not merely predictive loss.

For each firm-year (i, t), the forecast panel contains two model outputs:

$$\widehat{\mathbf{y}}_{i,t} = \left[\widehat{R}_{i,t}, \widehat{\sigma}_{i,t}\right], \tag{22}$$

where $\widehat{R}_{i,t}$ is predicted forward cumulative return and $\widehat{\sigma}_{i,t}$ is predicted forward annualized volatility. These predictions are generated for the same July-to-June target window used in the supervised learning problem.

9.2 Annual Signal Persistence and Monthly Rebalancing

The ST-GAT model produces annual forecasts rather than monthly forecasts. A prediction associated with risk year t is therefore treated as valid from July $t+1$ through June $t+2$. Within this 12-month forecast window, the signal ranking is held fixed unless a new annual prediction becomes available. Monthly rebalancing is nevertheless economically meaningful because it controls weight drift, updates investable eligibility based on available price data, realizes transaction costs, and maintains the intended long, short, and risk-budget exposures.

The timing convention is designed to avoid look-ahead bias. At month-end m, the strategy observes the active annual signal available at that date, forms target weights, incurs turnover-based transaction costs, and earns the return from month m to month $m+1$. Thus, the rebalance decision and realized holding-period return are separated by one monthly interval.

$$\mathbf{w}_m^{target} = f\left(\mathcal{S}_m, \mathcal{U}_m\right), \qquad R_{p,m+1} = \sum_{i \in \mathcal{U}_m} w_{i,m}^{target} R_{i,m+1} - \text{Cost}_m, \tag{23}$$

where $\mathcal{S}_m$ is the active annual signal set and $\mathcal{U}_m$ is the investable universe with valid next-month return observations.

9.3 Score Definitions and Portfolio Ranking

Because the ST-GAT produces both return and volatility forecasts, we evaluate three alternative scoring rules. The first uses predicted return directly:

$$\text{Score}_{i,t}^{ret} = \widehat{R}_{i,t}. \tag{24}$$

The second converts predictions into a risk-adjusted return score:

$$\text{Score}_{i,t}^{ret/vol} = \frac{\widehat{R}_{i,t}}{\max(\widehat{\sigma}_{i,t}, 0.10)}, \tag{25}$$

where the 10% volatility floor prevents unstable weights caused by very small predicted volatility values. The third score uses standardized return and volatility forecasts:

$$\text{Score}_{i,t}^{comp} = z(\widehat{R}_{i,t}) - 0.5z(\widehat{\sigma}_{i,t}). \tag{26}$$

This composite score rewards high predicted return while penalizing high predicted volatility, without imposing the stronger functional form of a return-to-volatility ratio.

9.4 Long-Only and Long-Short Portfolio Construction

For each score definition, two strategies are constructed. The first is a long-only top-quintile portfolio. At each rebalance date, stocks are ranked by the active score, and the top 20% of eligible firms are selected. Within the selected set, weights are scaled inversely by predicted volatility:

$$\tilde{w}_{i,m} = \frac{1/\max(\widehat{\sigma}_{i,t}, 0.10)}{\sum_{j \in \mathcal{Q}_m^{top}} 1/\max(\widehat{\sigma}_{j,t}, 0.10)}. \tag{27}$$

A 5% single-name cap is applied to reduce concentration risk, after which weights are renormalized to sum to one.

The second strategy is a long-short top-minus-bottom quintile portfolio. The top 20% of stocks by score are held long and the bottom 20% are held short. The long leg is allocated 50% gross exposure and the short leg is allocated -50% gross exposure:

$$\sum_{i \in \mathcal{Q}_m^{top}} w_{i,m} = 0.5, \qquad \sum_{i \in \mathcal{Q}_m^{bottom}} w_{i,m} = -0.5. \tag{28}$$

This design creates an approximately market-neutral cross-sectional ranking test, while still allowing non-zero beta because no explicit beta-neutral optimization is imposed.

9.5 Transaction Costs, Slippage, and Turnover

Implementation frictions are modeled using one-way turnover. Let $\mathbf{w}_m^{target}$ denote the target weights at rebalance month m and $\mathbf{w}_m^{pre}$ denote the drifted weights immediately before rebalancing. Turnover is:

$$\text{Turnover}_m = \sum_i \left| w_{i,m}^{target} - w_{i,m}^{pre} \right|. \tag{29}$$

The total one-way implementation cost is set to 15 bps, consisting of 10 bps transaction cost and 5 bps bid-ask/slippage allowance:

$$\text{Cost}_m = 0.0015 \times \text{Turnover}_m. \tag{30}$$

Net portfolio return is therefore:

$$R_{p,m+1}^{net} = R_{p,m+1}^{gross} - \text{Cost}_m. \tag{31}$$

This treatment penalizes both initial portfolio formation and subsequent monthly rebalancing, making the backtest more conservative than a frictionless ranking exercise.

9.6 Evaluation Metrics

Portfolio performance is evaluated using total return, annualized return, annualized volatility, Sharpe ratio, Sortino ratio, maximum drawdown, hit rate, average monthly turnover, alpha and beta against SPY, information ratio, and correlation with the benchmark. The Sharpe ratio is the standard mean-variance reward-to-risk measure, while the Sortino ratio focuses on downside risk (Sharpe, 1966; Sortino and Price, 1994).

9.7 Portfolio Performance Results

Table 8 reports the headline strategies for the metrics currently exported by the backtesting engine. The strongest long-short strategy ranks stocks by predicted return only, rather than by predicted return divided by predicted volatility. This finding is important: although predicted volatility improves the forecasting task, using volatility directly in the denominator can over-penalize high-return opportunities. The return-only signal is therefore the cleanest alpha ranking.

Table 8: Key portfolio backtest results for Meso ST-GAT signals. Dashes indicate metrics not exported in the current summary table.

Strategy	**Ann. Return**	**Ann. Vol.**	**Sharpe**	**Sortino**	**Max DD**
GAT Long-Short Return-Only	7.93%	7.47%	**1.06**	**2.01**	**-6.90%**
GAT Long-Only Return-Only	**22.71%**	–	1.04	–	-13.99%
SPY Benchmark	10.74%	–	0.65	–	-23.93%

The GAT long-short return-only portfolio achieves 7.93% annualized return with only 7.47% annualized volatility, producing a Sharpe ratio of 1.06 and Sortino ratio of 2.01. Its maximum drawdown is -6.90%, substantially lower than SPY's -23.93%. Its beta against SPY is approximately 0.24, indicating that the strategy is not merely a levered market exposure. The GAT long-only return-only portfolio also performs strongly, generating 22.71% annualized return, a Sharpe ratio of 1.04, maximum drawdown of -13.99%, and annualized alpha of approximately 9.85% against SPY.

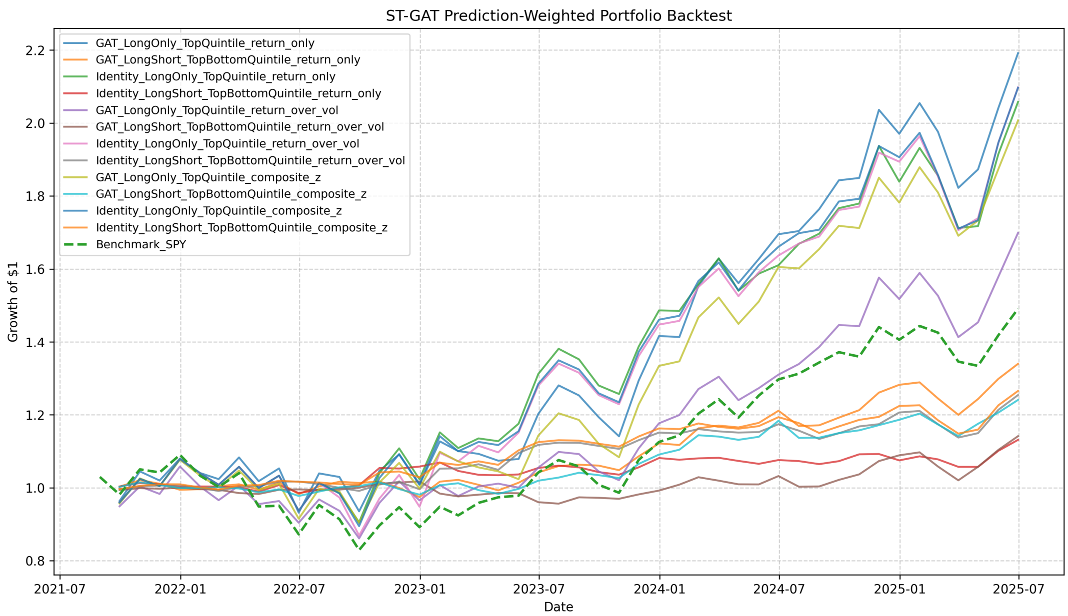

Figure 11: Monthly rebalanced ST-GAT and identity-baseline portfolio equity curves.

9.8 Score Ablation Interpretation

The score ablation shows that the ST-GAT return-only ranking is more effective than the return-over-volatility ranking. This does not contradict the volatility-forecasting improvement documented in Section 8. Rather, it indicates that the predicted volatility output is useful for understanding and controlling risk, but the most direct source of cross-sectional alpha is the predicted return ranking. The composite score performs between the return-only and return-over-volatility specifications, suggesting that moderate volatility penalization is preferable to mechanically dividing by predicted volatility.

This result also clarifies the difference between forecast evaluation and trading evaluation. RMSE and MAE measure point-forecast accuracy, while a portfolio primarily depends on the ordinal separation of expected winners and losers. Therefore, the portfolio section emphasizes Sharpe ratio, drawdown, beta, turnover, and top-minus-bottom spread in addition to predictive metrics.

9.9 Top-Bottom Spread Diagnostic

To directly test whether the predicted rankings separate future winners from losers, we compute the monthly average return of the top quintile minus the bottom quintile. This diagnostic is model-agnostic: it does not depend on inverse-volatility weights, portfolio caps, or transaction-cost assumptions. It simply asks whether stocks ranked highly by the signal outperform stocks ranked poorly by the signal in the next month.

Table 9: Top-minus-bottom spread diagnostics for return-only signals.

Signal	**Annualized Spread**	***t*-stat**	***p*-value**	**Positive Spread Rate**
ST-GAT Return-Only	**17.69%**	**2.18**	**0.034**	**65.2%**
Identity Return-Only	6.62%	1.34	0.189	–

The ST-GAT return-only signal produces a statistically significant 17.69% annualized top-minus-bottom spread, with $t = 2.18$ and $p = 0.034$. The spread is positive in 65.2% of evaluated months. By contrast, the identity baseline return-only signal produces a smaller 6.62% annualized spread with $t = 1.34$ and $p = 0.189$, which is not statistically significant. This confirms that the Meso textual-risk topology improves the economic separability of future winners and losers beyond what can be achieved using each equity's own features alone.

9.10 Portfolio-Level Implications

The portfolio extension supports three conclusions. First, the Meso ST-GAT signal is economically meaningful: it improves not only statistical forecast metrics but also tradable cross-sectional ranking. Second, the best-performing allocation uses predicted return directly, showing that the alpha signal is primarily embedded in the return forecast rather than the return-volatility ratio. Third, monthly rebalancing is compatible with annual predictions when the signal is interpreted as a persistent 12-month forecast and rebalancing is used for risk control and execution rather than signal refresh. These findings motivate the use of LLM-derived semantic risk topology as an input to systematic equity selection.

10 LIMITATIONS

This study identifies several structural and methodological limitations within the proposed Spatio-Temporal Graph Attention Network, the underlying NLP pipeline, and the portfolio extension. First, there is an inherent frequency mismatch between annual 10-K risk disclosures and continuous market dynamics. In the core ST-GAT architecture, this is handled by representing textual risk as an annual cross-sectional graph snapshot. This design is appropriate for one-year-ahead return and volatility forecasting, but it cannot capture intra-year fundamental shocks unless additional higher-frequency textual or event data are incorporated.

Second, the supplemental DAV experiment illustrates the difficulty of mapping annual risk disclosures into daily volatility models. Forward-filling annual risk scores into daily EGARCH-X equations often introduced statistical noise, with positive BIC improvements concentrated only in highly specific cases such as patent litigation in biopharmaceuticals or regulatory risk in energy services. This limitation motivates the graph-based cross-sectional architecture, but it also cautions against interpreting annual textual risk scores as high-frequency trading signals.

Third, the predictive efficacy of the spatial graph topology is bounded by disclosure quality. Firms with boilerplate or near-zero-drift disclosures can introduce persistent but weakly informative links into the adjacency matrix. The strict identity-matrix ablation and the Macro-versus-Meso comparison partially address this issue, but a future implementation should explicitly filter low-drift firms or down-weight stale disclosures when constructing graph topology.

Fourth, the dynamic taxonomy framework relies on a fixed semantic-deviation threshold ($\theta = 0.45$). Although the validation experiment based on 2022 and 2024 data shows improved coverage, a full longitudinal sensitivity analysis would be needed to assess the stability of this threshold across the entire 2006–2024 period. There is also residual risk of LLM-induced semantic drift when categories are repeatedly added, merged, or reorganized.

Fifth, secondary validations such as the Spatial Autoregressive network analysis remain constrained by extreme matrix sparsity and small sample sizes. This highlights the difficulty of modeling continuous market dynamics strictly through annualized semantic linkages.

Finally, the portfolio backtest remains an academic backtest rather than a live-trading system. Transaction costs and slippage are included, but borrow constraints, short-sale availability, execution delay, corporate actions, liquidity constraints, and a fully survivorship-bias-free reconstruction of the S&P 500 universe remain only partially addressed. These limitations should be resolved before interpreting the strategy as deployable capital.

11 CONCLUSION & NEXT STEPS

This study develops a comprehensive, LLM-driven framework for transforming lengthy and unstructured SEC Item 1A risk disclosures into quantitative forecasting and portfolio-construction signals. The pipeline extracts risk paragraphs from 10-K filings, maps them into a hierarchical macro–meso taxonomy, converts firm-year disclosures into normalized risk exposure vectors, and uses those vectors to construct dynamic semantic peer graphs.

The central modeling contribution is a topology-only Spatio-Temporal Graph Attention Network with a strict identity-matrix ablation. The full model uses $\mathbf{A}_t^{full} = \mathbf{I} + \mathbf{A}_t^{risk}$, while the baseline uses $\mathbf{A}_t^{base} = \mathbf{I}$. This design isolates the incremental value of textual peer topology beyond each equity's own financial features. In the 2021–2024 OOS period, the Meso ST-GAT outperforms the identity baseline across return and volatility forecasting metrics, improving averaged RMSE from 0.3345 to 0.3001 and averaged Spearman rank from -0.0139 to 0.1701. The strongest improvement occurs in volatility forecasting, where RMSE falls from 0.2240 to 0.1565 and Spearman rank improves from -0.1351 to 0.1770.

Graph diagnostics further clarify why semantic resolution matters. The Macro graph is relatively dense during the OOS period, retaining roughly 3,900–4,460 off-diagonal directed edges per year, while the Meso graph retains only 38–118 edges. The superior performance of the sparse Meso graph supports the interpretation that granular textual-risk categories form high-confidence semantic peer links, whereas broad Macro categories can over-propagate generic disclosure noise.

The portfolio extension demonstrates that the Meso ST-GAT forecasts are not only statistically useful but also economically meaningful. Annual predictions are held fixed over their July-to-June forecast window and portfolios are rebalanced monthly to control drift, turnover, and implementation costs. The strongest long-short strategy ranks stocks by predicted return, holds the top quintile long and bottom quintile short, and achieves a Sharpe ratio of 1.06, Sortino ratio of 2.01, maximum drawdown of -6.90%, and beta of 0.24 against SPY. The top-minus-bottom spread diagnostic is also statistically significant, with a 17.69% annualized spread, $t = 2.18$, and $p = 0.034$, compared with an insignificant identity-baseline return-only spread.

Supplemental DAV, Fama-MacBeth, and SAR analyses provide additional economic context. DAV results show that annual textual risks are difficult to use in isolated high-frequency volatility models except in highly specific fundamental cases. Fama-MacBeth regressions show that selected Meso-level vulnerabilities carry statistically meaningful cross-sectional risk premiums, while SAR analysis suggests preliminary evidence of textual-peer volatility spillovers. Together, these results support the broader conclusion that granular LLM-derived risk signals are most powerful when used as structured cross-sectional topology rather than as undifferentiated text sentiment.

Future work should extend the framework in four directions. First, historical constituent reconstruction and delisting-aware price histories should be improved to further mitigate survivorship bias. Second, disclosure staleness and boilerplate intensity should be integrated directly into edge weighting or edge filtering. Third, monthly or quarterly textual/event signals could be added to support genuinely higher-frequency forecasts. Finally, the portfolio engine should be expanded to include borrow constraints, beta-neutral optimization, liquidity-aware execution, and transaction-cost stress testing.

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A Econometric Baseline Specification (DAV Model)

To establish a rigorous traditional econometric baseline for textual risk integration, we evaluated a Disclosure Augmented Volatility (DAV) model. The DAV framework extends a standard Exponential GARCH (EGARCH) model by introducing textual risk scores as exogenous variables (EGARCH-X). This comparative econometric analysis evaluates standard GARCH(1,1), EGARCH(1,1), and the proposed DAV model.

A.1 Data Sources and Preprocessing

The model evaluates the 2020–2025 temporal regime. Daily adjusted closing prices were sourced via Yahoo Finance to calculate realized continuous returns (r_t). The textual risk signals utilized the granular Meso-level risk matrices derived from the 10-K filings. To align the annual reporting frequency of 10-Ks with daily trading data, the annual risk scores were forward-filled, creating a daily step-function vector for the risk level (S_t) and a discrete difference vector for the risk shock (dS_t).

A.2 Mathematical Formulation

The DAV model is specified as an EGARCH(1,1)-X process. Unlike standard GARCH, the EGARCH framework models the natural logarithm of the conditional variance ($\ln(h_t)$), ensuring positive volatility without enforcing artificial non-negativity constraints on the coefficients. It also captures the asymmetric "leverage effect" of negative returns.

The textual risk factors are embedded directly into the conditional variance equation:

$$\ln(h_t) = \omega + \beta \ln(h_{t-1}) + \alpha(|z_{t-1}| - \mathbf{E}[|z|]) + \gamma z_{t-1} + \delta_1 S_t + \delta_2 dS_t \tag{32}$$

Where:

- h_t is the conditional variance at time t.
- z_{t-1} is the standardized residual $\frac{\epsilon_{t-1}}{\sqrt{h_{t-1}}}$.
- γ captures the asymmetric leverage effect of return shocks.
- δ_1 measures the persistent "Level Effect" of the firm's textual risk score.
- δ_2 captures the transient "Shock Effect" representing year-over-year changes in disclosure.

The models were optimized using the L-BFGS-B algorithm to minimize the negative log-likelihood of the Gaussian distribution. Model selection was governed by the Bayesian Information Criterion (BIC), which severely penalizes models for adding uninformative parameters.

A.3 Results and Discussion

The empirical results of the DAV benchmark highlight a fundamental limitation of isolated time-series econometrics when processing natural language signals. For the vast majority of the equity universe, the inclusion of textual risk levels and shocks resulted in a negative BIC improvement. The standard EGARCH(1,1) model frequently outperformed the DAV model, indicating that for isolated single-asset forecasting, textual risk parameters often introduce more statistical noise than predictive alpha.

However, the DAV model demonstrated isolated pockets of exceptional predictive power when a firm's volatility was overwhelmingly driven by specific, granular Meso-risks:

- **REGN (Regeneron Pharmaceuticals):** The model recorded a substantial BIC improvement of 56.80 when conditioned on *Intellectual Property Challenges and Litigation*. This aligns with the reality that biopharmaceutical volatility is heavily gated by patent cliffs and IP litigation, making this specific textual disclosure a highly potent exogenous variable.
- **SLB (Schlumberger):** Achieved a substantial BIC improvement of 51.85 when conditioned on *Regulatory and Legal Action Risk*. As a major oilfield services company, SLB's realized volatility is deeply intertwined with environmental regulations and cross-border compliance shocks.
- **V (Visa):** Saw a substantial BIC improvement of 20.07 driven by *Integration and Execution Risks* related to strategic transactions, reflecting the market's sensitivity to Visa's M&A pipeline and fintech integration hurdles.

The visual convergence of these specific improvements can be observed in Figure 12.

Conclusion: The broader failure of the DAV model across the general market, juxtaposed with its success in highly specific fundamental cases, mathematically justifies the shift toward Spatio-Temporal Graph Neural Networks. Traditional EGARCH-X models fail because they cannot contextualize risk; they ignore the peer-to-peer contagion effects of risk factors. The ST-GAT architecture resolves this by routing risk dynamically across a connected topology, rather than forcing it into an isolated time-series equation.

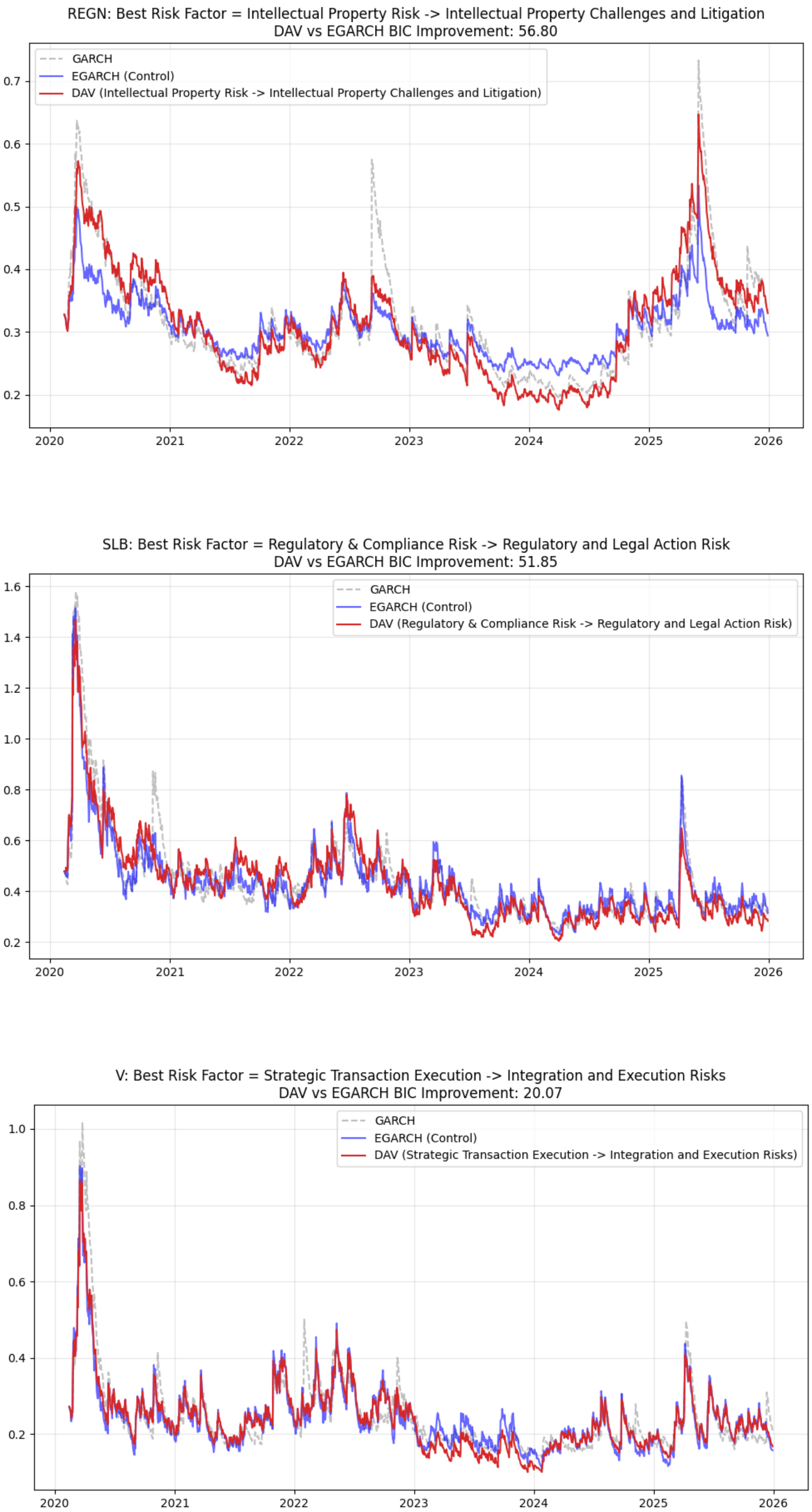

Figure 12: DAV vs EGARCH Volatility Forecasts illustrating positive BIC improvements for REGN, SLB, and V.

B Econometric Baseline Specification (Fama-MacBeth Regressions)

To rigorously evaluate whether the textually derived Meso-level risk vulnerabilities carry a statistically significant risk premium in the equity markets, we implement a standard Fama-MacBeth (1973) two-stage regression framework. Unlike the DAV model which analyzes isolated time-series volatility, the Fama-MacBeth methodology cross-sectionally evaluates whether exposure to specific natural language risk factors systematically prices forward expected returns.

B.1 Data Sources and Mathematical Formulation

The analysis utilizes the granular Meso-level risk scores extracted from the annual 10-K filings. To align with standard asset pricing literature, these annual risk exposures are held constant over the subsequent 12 months and regressed against one-month forward continuous returns ($r_{i,t+1}$), calculated using adjusted closing prices from Yahoo Finance over the 2020–2025 temporal regime. To overcome the high dimensionality and sparsity of the Meso-risk matrix (where the number of textual parameters K exceeds the cross-sectional equity universe N for specific months), the models are specified as univariate regressions.

Stage 1: Cross-Sectional Regressions
For each month t, a cross-sectional Ordinary Least Squares (OLS) regression is estimated across all active firms i:

$$r_{i,t+1} = \alpha_t + \lambda_{k,t} S_{i,k,t} + \epsilon_{i,t+1} \quad \text{for } i = 1, \ldots, N_t \tag{33}$$

Where $S_{i,k,t}$ is the textual exposure of firm i to Meso-risk factor k at time t, and $\lambda_{k,t}$ is the estimated cross-sectional risk premium.

Stage 2: Time-Series Averaging
The final risk premium ($\hat{\lambda}_k$) is computed as the time-series average of the monthly estimates:

$$\hat{\lambda}_k = \frac{1}{T} \sum_{t=1}^{T} \hat{\lambda}_{k,t} \tag{34}$$

Standard errors are corrected using the Newey-West estimator with a 3-month lag structure, producing robust t-statistics to control for autocorrelation and heteroskedasticity.

B.2 Empirical Results and Economic Analysis

Table 10 and Figure 13 detail the top 10 most statistically significant Meso-level risk premiums identified during the 2020–2025 regime.

Table 10: Fama-MacBeth Univariate Regression Results: Top 10 Meso-Risk Factors

Meso-Risk Factor (10-K NLP Extraction)	**Risk Premium ($\hat{\lambda}$)**	**Newey-West SE**	**t-statistic**	**p-value**
Pandemic and Public Health Crisis	-0.0867	0.0348	-2.49	0.0126
Strategic Initiative Execution & Integration Risk	-0.0899	0.0374	-2.40	0.0162
Regulatory and Compliance Risks	-0.0455	0.0239	-1.90	0.0575
Geopolitical Conflict and Sanctions Risk	-0.4008	0.2130	-1.88	0.0598
Equity Structure and Capital Market Actions	0.2482	0.1328	1.87	0.0616
Public Health Crises & Pandemics (Variant)	-0.0681	0.0377	-1.81	0.0707
Accounting Standards & Policy Compliance	0.0879	0.0513	1.71	0.0869
Inventory and Demand Volatility	0.0782	0.0462	1.69	0.0904
Digital & Channel Evolution	-0.0263	0.0157	-1.67	0.0950
Patent Exclusivity Loss	-0.2766	0.1688	-1.64	0.1012

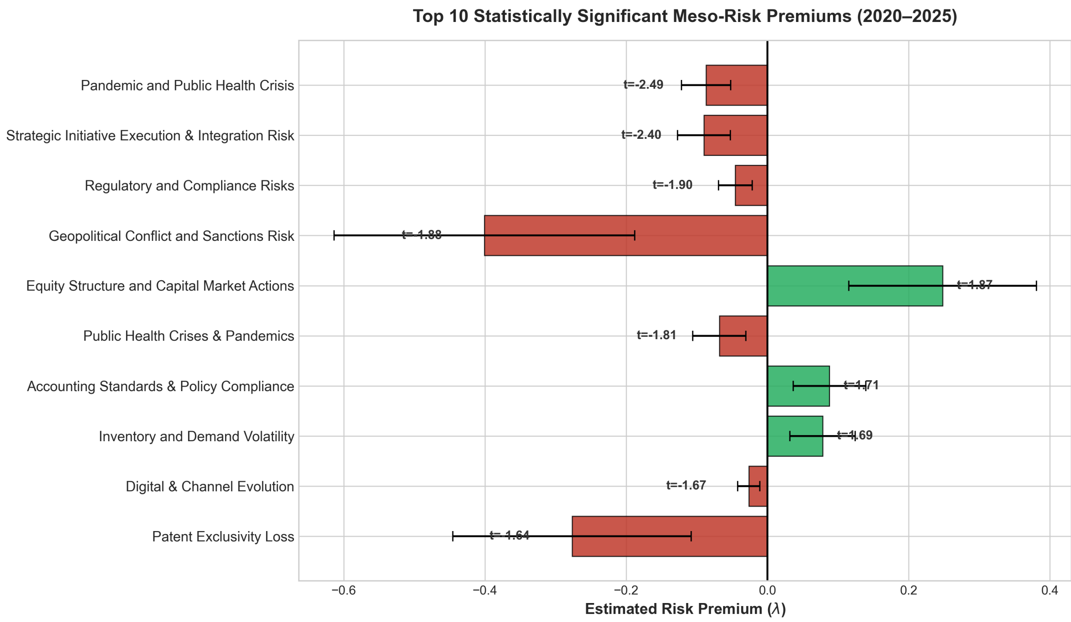

Figure 13: Cross-sectional risk premiums ($\hat{\lambda}$) for the top 10 significant textual risk factors. Error bars represent Newey-West standard errors. Negative values indicate the risk acted as a drag on forward returns.

The empirical evidence reveals that the equity market systematically prices several highly specific vulnerabilities, particularly those aligned with systemic shocks during the out-of-sample window. Most notably, exposure to *Pandemic and Public Health Crises* yielded a statistically significant negative risk premium ($t = -2.49$). A similarly significant negative premium was observed for *Strategic Initiative Execution Risk* ($t = -2.40$).

The negative coefficients indicate that high textual exposure to these specific fundamental shocks acted as a severe drag on forward expected returns. Investors dynamically discounted the valuations of equities that heavily disclosed these vulnerabilities, demanding a lower price to hold the asset, which resulted in negative realized returns over the period. Interestingly, structural risks such as *Equity Structure and Capital Market Actions* carried a positive premium ($t = 1.87$), indicating that firms undertaking structural capitalization shifts compensated investors with higher subsequent returns.

Conversely, idiosyncratic corporate boilerplate such as *Dependence on Key Collaborators and Products* completely failed to cross the significance threshold ($t = -0.01$, $p = 0.99$). The market effectively treats these ubiquitous disclosures as unpriced noise.

These findings mathematically reinforce the core necessity of the Spatio-Temporal Graph Attention Network (ST-GAT) presented in the main text. Because the market only prices highly specific vectors of vulnerability, textual NLP models cannot treat all language equally; they must utilize granular, Meso-level semantic routing to isolate the true vectors of risk contagion from the surrounding regulatory noise.

C Risk Contagion Network Analysis

C.1 Objective and Motivation

To validate the practical utility of the evolved taxonomy, we examine whether textual risk disclosure similarities in 10-K filings can predict risk contagion—specifically the synchronization of volatility—within the stock market. We hypothesize that firms sharing similar "risk signatures" in their disclosures (textual peers) will exhibit correlated stock price volatility.

C.2 Network Construction Methodology

We transform raw 10-K disclosures into a quantitative risk spillover model through a multi-stage pipeline:

C.2.1 Risk Exposure Vectors and Similarity

For each firm i at time t, we define a Risk Exposure Vector $\mathbf{R}_{i,t}$:

$$\mathbf{R}_{i,t} = (w_{i,t,1}, w_{i,t,2}, \ldots, w_{i,t,K}) \tag{35}$$

where $w_{i,t,k}$ represents the normalized count of risk factors assigned to meso-category k. The similarity between any two firms i and j is computed using cosine similarity:

$$\text{sim}(\mathbf{R}_i, \mathbf{R}_j) = \frac{\mathbf{R}_i \cdot \mathbf{R}_j}{\|\mathbf{R}_i\| \|\mathbf{R}_j\|} \tag{36}$$

C.2.2 Spatial Weights Matrix (W)

To define the network topology, we construct a Binary Adjacency matrix $\mathbf{A}_{i,j}$ based on a similarity threshold τ:

$$\mathbf{A}_{i,j} = \begin{cases} 1 & \text{if } \text{sim}(\mathbf{R}_i, \mathbf{R}_j) > \tau \\ 0 & \text{otherwise} \end{cases} \tag{37}$$

A value of 1 indicates a "Risk Peer" relationship exists. This is then row-normalized to create the Spatial Weights Matrix $\mathbf{W}_{i,j}$:

$$\mathbf{W}_{i,j} = \frac{\mathbf{A}_{i,j}}{\sum_{j=1}^{n} \mathbf{A}_{i,j}} \tag{38}$$

C.3 Spatial Autoregressive (SAR) Framework

We integrate the NLP-derived risk network with 2023 market price data using a Spatial Autoregressive (SAR) model:

$$\mathbf{y} = \rho \mathbf{W} \mathbf{y} + \boldsymbol{\alpha} + \boldsymbol{\epsilon} \tag{39}$$

Where:

- $\mathbf{y}$: The dependent variable, represented by the **2023 Annualized Realized Volatility** (calculated from daily log returns via the Stooq API).
- $\mathbf{W}$: The **Spatial Weights Matrix** derived from the evolved taxonomy.
- ρ (Rho): The **Spatial Lag coefficient**, measuring the "contagion" effect—how much a firm's volatility is influenced by the volatility of its textual peers.
- $\boldsymbol{\alpha}$: The **baseline idiosyncratic volatility** of a firm, capturing the expected level of risk that remains constant regardless of the spillover effects or contagion within the textual risk network.

C.4 Preliminary Results and Validation

Validation was completed with a preliminary sample of 67 firms. The results indicate a positive risk spillover effect between textual neighbors:

- **Spatial Lag (ρ):** 0.0916. This positive coefficient indicates a correlation in risk levels across the network, suggesting that when a firm's textual peers experience high volatility, the firm is likely to experience an increase in volatility as well.
- **P-value:** 0.1048. While slightly above the traditional 5% significance level, this indicates a marginal trend in the predicted direction given the small sample size.
- **R-squared:** 4.1%. This suggests that 10-K text alone captures a meaningful portion of cross-sectional volatility variation, even without traditional financial metrics (e.g., leverage, size).

The findings confirm that the dynamic taxonomy successfully identifies "Risk Peers" that are economically linked, providing a novel tool for systemic risk monitoring and portfolio diversification.

D Representative Risk Taxonomy (Sample)

Due to the high dimensionality and breadth of the evolved risk taxonomy, Table 11 presents a representative excerpt of the system's output. This sample illustrates the hierarchical mapping between established **Macro Categories** and the more granular **Meso Categories** (Micro-level).

Table 11: Extracted Risk Taxonomy: Macro and Micro Categories

Macro Category	**Micro Categories**
Key Person and Succession Risk	Executive Leadership and Critical Function Dependency Risk, Executive Leadership Dependency and Transition Risk
Commodity Price & Supply Chain Volatility	Raw Material Input Cost Inflation, Human Capital & Labor Market Disruption
Strategic and Operational Market Vulnerabilities	Intellectual Property and Innovation Competition, Regulatory and Reimbursement Environment
Contractual & Regulatory Performance Risk	U.S. Government Funding & Procurement Dependency Risk, Government Contracting & Performance Risk
Data Privacy & Cybersecurity Regulatory Compliance Risk	Regulatory Compliance & Legal Liability Risk for Personal Data, Cross-Jurisdictional Data Privacy Law Compliance Risk
Human Capital Acquisition & Retention Risk	Key Personnel Dependency & Talent Competition Risk, Talent Acquisition & Retention Risk
Human Capital Management & Organizational Culture	Talent Development & Learning Systems, Diversity, Equity, Inclusion & Belonging (DEIB) Strategy & Goals
Regulatory, Legal, and Tax Compliance Volatility	Tax Law and Liability Volatility, Tax Policy and Obligation Volatility, Tax Provision and Contingency Volatility
Capital Allocation and Shareholder Returns Governance	Structural and Regulatory Constraints on Capital Deployment, Dividend Policy Discretion and Volatility Risk, Stock Price Volatility and Market Perception Risk, Corporate Governance and Controlling Shareholder Risks
Healthcare Policy, Pricing, and Reimbursement Risk	Government Program Contracting & Compliance Risk, Reimbursement and Pricing Pressure Risk
Legal, Regulatory, and Insurance Risk Exposure	Insurance Adequacy & Indemnification Risk, Litigation and Legal Contingency Risk
Strategic & Market Environment Risk	Global Regulatory & Food Safety Compliance Risk, Competitive & Market Dynamics Risk

Continued on next page

Table 11 – continued from previous page

Macro Category	**Micro Categories**
Intellectual Property Rights & Exclusivity Management	Patent Portfolio Viability & Litigation Exposure, Third-Party IP Infringement Claims & Defense Costs, Loss of Exclusivity and Generic/Biosimilar Competition, Third-Party Intellectual Property Dependencies & Licensing, Intellectual Property Infringement & Defense
Product Commercialization and Market Dynamics	Manufacturing, Supply Chain, and Operational Capacity, Clinical Development & Safety Outcomes, Dependency on Strategic Collaborations and Market Exclusivity
Enterprise Risk Governance & Disclosure Framework	Regulatory & Disclosure Evolution Risk, Risk Factor Disclosure Framework
International Trade & Geopolitical Exposure	Cross-Border Regulatory & Compliance Complexity, Global Supply Chain & Market Access Disruption
Counterparty and Credit Risk	Investment Portfolio Valuation and Impairment Risk, Credit Rating Downgrade Risk, Debt Refinancing and Liquidity Risk, Regulatory Capital & Liquidity Compliance Risk, Customer and Counterparty Credit Exposure
Human Capital & Workforce Management	Labor Cost & Workforce Stability, Talent Acquisition, Retention & Succession, Pandemic-Induced Operational & Economic Disruption, Workforce Health, Safety, & Operational Continuity
Strategic Transaction & Integration Execution Risk	Acquisition Integration & Synergy Realization Risk, Post-Transaction Integration & Synergy Realization Risk
Climate & Environmental Disruption Risk	Physical Operational Disruption from Natural Hazards, Transition & Regulatory Compliance Risk
Third-Party and Supply Chain Dependency	Manufacturing & Production Disruption, Third-Party Operational and Performance Risk, Supplier Performance and Continuity Risk
Product & Brand Integrity and Market Acceptance	Product Safety, Quality, and Public Health Perception, Consumer Preference and Health Perception Risk, Bottler Partnership and System Integrity, Competitive Positioning and Market Dynamics
Business Model and Market Evolution Risk	Product Portfolio and Technology Lifecycle Risk, Competitive Pressure and Market Positioning Risk